



Lab 2 – Testing ASP.NET Core APIs Using Swagger and Postman

Learning Objectives

By the end of this lab, students will be able to:

- Understand why APIs must be tested without a frontend
- Use Swagger UI to test API endpoints
- Send GET and POST requests using Swagger
- Use Postman to send HTTP requests
- Inspect JSON responses returned from an API

Part 1 – Enabling Swagger

Modify Program.cs to enable Swagger support.

```
var builder = WebApplication.CreateBuilder(args);
```

```
// Enable OpenAPI / Swagger
```

```
builder.Services.AddEndpointsApiExplorer();
```

```
builder.Services.AddSwaggerGen();
```

```
var app = builder.Build();
```

```
if (app.Environment.IsDevelopment())
```

```
{
```

```
    app.UseSwagger();
```

```
    app.UseSwaggerUI();
```

```
}
```

```
app.MapGet("/greet", (string name) =>
```

```
{
```

```
    return new
```

```
{
```

```
        English = $"Hello {name}!",
```

```
        Arabic = $"مرحبا {name}!"
```

```
    };
```

```
});
```

```
app.MapPost("/greetpost", (NameRequest request) =>
```

```
{
```

[WebII lab sheet]

return new
[2nd Semester] [2025 - 2026]

[Lab 4]

Instructor

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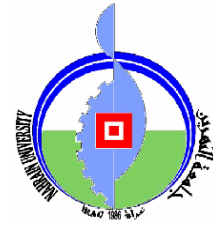
[March 8, 2026]

Email

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Hours

[2 Hours]



});

app.Run();

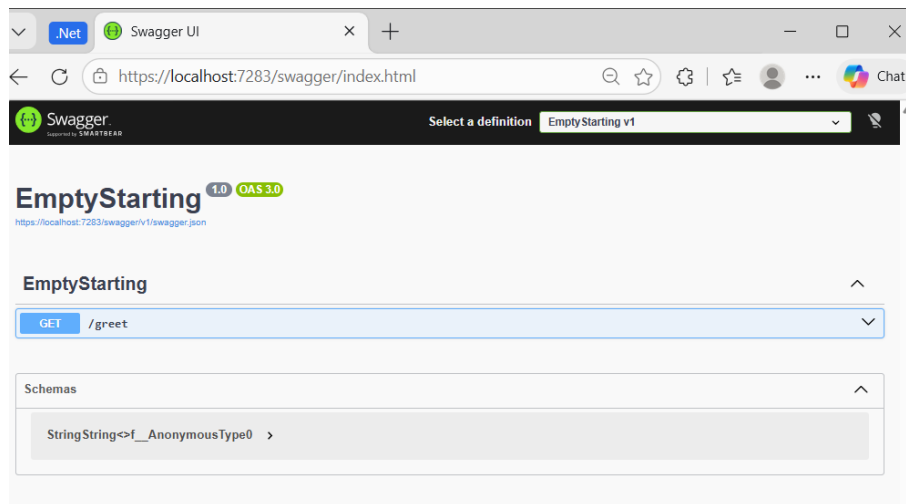
```
public class NameRequest
{
    public string Name { get; set; }
}
```

Part 2 – Running Swagger

Run the application and open the following URL in your browser:

<https://localhost:xxxx/swagger>

You will see a list of API endpoints that can be tested directly.



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Swagger UI

https://localhost:7283/swagger/index.html

Select a definition Empty Starting v1

EmptyStarting ^{1.0} OAS 3.0

https://localhost:7283/swagger/v1/swagger.json

EmptyStarting

GET /greet

Try it out

Parameters

Name	Description
name * required	
string	name
(query)	

Responses

Code	Description	Links
200	OK	No links

Media type: application/json

Controls: Accept header

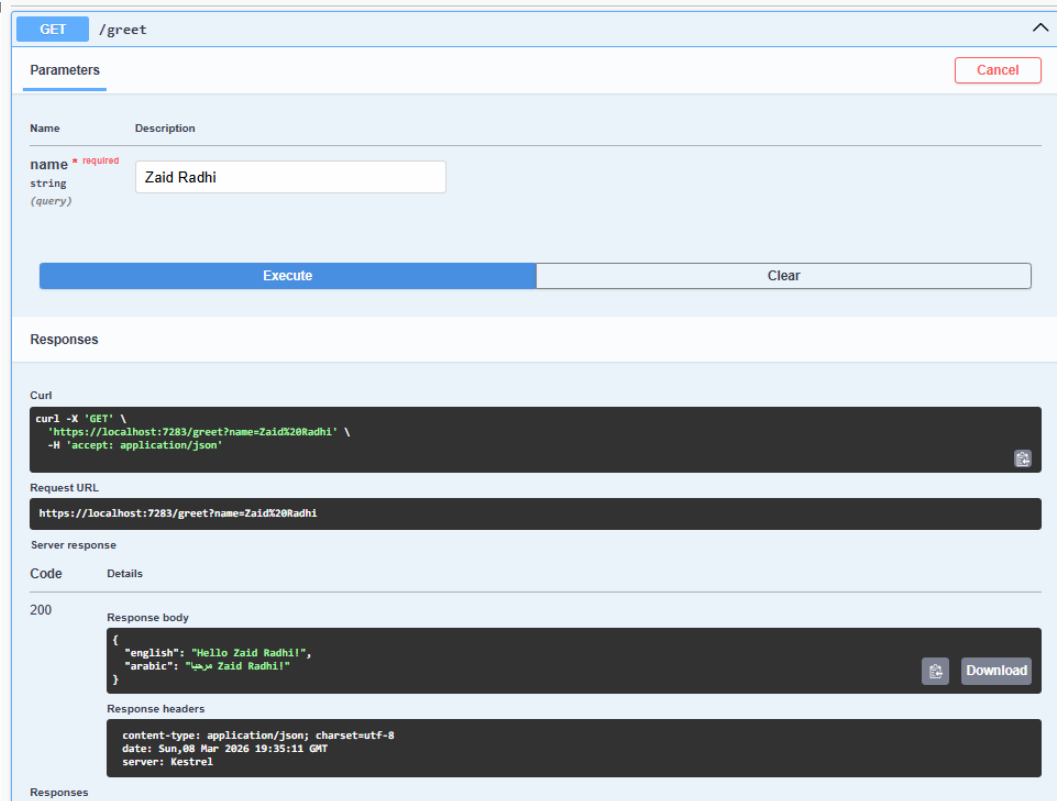
Example Value | Schema

```
{  "english": "string",  "arabic": "string"}
```



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EmptyStarting



The image shows the Swagger UI for the EmptyStarting API. The top bar indicates the API name 'EmptyStarting' with version '1.0' and 'OAS 3.0'. Below this, the URL 'https://localhost:7283/swagger/v1/swagger.json' is displayed. The main section is titled 'EmptyStarting' and shows the 'GET /greet' endpoint. Under the 'Parameters' tab, there is a single parameter 'name' of type 'string' (query), marked as 'required'. A text input field contains the value 'Zaid Radhi'. Below the parameters, there are 'Execute' and 'Clear' buttons. The 'Responses' section shows a '200' status code. The 'Response body' is a JSON object: { "english": "Hello Zaid Radhi!", "arabic": "مرحباً Zaid Radhi!" }. The 'Response headers' section shows: content-type: application/json; charset=utf-8, date: Sun, 08 Mar 2026 19:35:11 GMT, and server: Kestrel. A 'Curl' section shows the command: curl -X 'GET' \ 'https://localhost:7283/greet?name=Zaid%20Radhi' \ -H 'accept: application/json'. A 'Request URL' section shows: https://localhost:7283/greet?name=Zaid%20Radhi. A 'Server response' section shows the status code 200. A 'Details' section shows the response body and headers. A 'Download' button is present next to the response body.

GET /greet

Parameters

Cancel

Name	Description
name * required string (query)	Zaid Radhi

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'https://localhost:7283/greet?name=Zaid%20Radhi' \
-H 'accept: application/json'
```

Request URL

```
https://localhost:7283/greet?name=Zaid%20Radhi
```

Server response

Code	Details
200	Response body <pre>{ "english": "Hello Zaid Radhi!", "arabic": "مرحباً Zaid Radhi!" }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Sun, 08 Mar 2026 19:35:11 GMT server: Kestrel</pre>

Download

Testing the GET Endpoint

1. Expand GET /greet
2. Click 'Try it out'
3. Enter a name
4. Click Execute

Testing the POST Endpoint

1. Expand POST /greetpost
2. Click 'Try it out'
3. Enter JSON body:
{ "name": "Ali" }
4. Click Execute

Part 3 – Testing with Postman

Postman is a professional API testing tool widely used in industry.

GET Request in Postman

[WebII lab sheet]

Method: GET

[2nd Semester] [2025 - 2026]

URL: `https://localhost:xxxx/greet?name=Ali`

[Lab 4]

Instructor

Click Send to view the response.

[March 8, 2026]

Email

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POST Request in Postman

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Hours

[2 Hours]



Method: POST

URL: `https://localhost:xxxx/greetpost`

Body raw JSON:

```
{ "name": "Ali" }
```

Student Task

Modify the API so that the response also returns the length of the name.

Example:

```
{ "english": "Hello Ali!", "arabic": "مرحبا Ali!", "nameLength": 3 }
```